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Jellyfish-inspired multilayered responsive hydrogels *via* micelle-confined photopolymerization for underwater communication and emergency rescue

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Hydrogels offer considerable promise as flexible sensors. However, maintaining their stable performance in complex underwater environments characterized by high pressure, high salinity, and temperature fluctuations remains a significant challenge. Inspired by the dynamic adaptive capabilities of jellyfish, this study reports a biomimetic composite conductive hydrogel with multi-level environmental responsiveness, synthesized *via* the copolymerization of *N*-isopropylacrylamide, 2-acrylamido-2-methylpropanesulfonic acid, and octadecyl methacrylate (OMA), incorporating color-changing microcapsules through a cetyltrimethylammonium bromide (CTAB) micelle-confined polymerization strategy. The long alkyl chains of OMA promote hydrophobic association networks inside CTAB micelles, endowing the hydrogel with both high strength and exceptional stretchability (fracture strength of 690 kPa; elongation at break 1370%). This hydrogel achieves multi-parameter, synchronized responses to complex marine conditions: strong temperature sensitivity (−15.344%/°C), an electrical response to water-depth pressure (7.13 kPa^{-1}, corresponding to 0–500 m water depth), stable tensile responsiveness, and reversible color changes visible to the naked eye. Building on these capabilities, the hydrogel can monitor human motion in conventional settings and also enable underwater sensing and Morse-code-based information transmission, demonstrating substantial potential for applications for underwater communication and emergency rescue applications.

New concepts

Inspired by the dynamic adaptive capabilities of jellyfish, we developed a robust biomimetic composite conductive hydrogel engineered specifically for complex marine environments. It exhibits exceptional stretchability and strength. Featuring multi-parameter responsiveness to temperature, water-depth pressure, and strain, alongside visible color changes, this hydrogel enables reliable underwater human motion monitoring, Morse-code communication, and emergency rescue sensing.

1. Introduction

Underwater environments pose unique challenges for sensing technologies that differ significantly from those encountered in air. While sensors in air primarily need to cope with fluctuations in humidity,[1,2] temperature,[3,4] and gas composition,[5,6] underwater environments—particularly in the deep sea—feature complex interactions among multiple factors. Salinity,[7] hydrostatic pressure,[8–11] and temperature interact synergistically,[12] influencing sensor performance in ways not seen in atmospheric settings. Consequently, creating underwater sensors that are both stable and highly sensitive, capable of simultaneously withstanding high pressure, high salinity, and variable temperatures, has become a critical focus in marine exploration technology. Recent advances in underwater flexible sensors illustrate these efforts. To adapt to complex aquatic environments, recent research has predominantly focused on developing specific functional properties—such as underwater adhesion,[13] self-healing,[14] or extreme mechanical strength.[15] However, this specialization often leads to a lack of multimodal sensing versatility. For instance, Guo *et al.*[16] successfully incorporated hydrophobic segments into a dynamically crosslinked network to achieve exceptional underwater self-healing capability. However, this design is limited to a single-modal electrical output, lacking the comprehensive sensing capabilities required for complex external environments. Similarly, Ashis *et al.*[17] designed a muscle-inspired anisotropic hydrogel sensor based on pre-stretching and metal-ion coordination. Although it demonstrates excellent underwater anti-expansion performance, it still fails to

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provide intuitive multimodal feedback in low-visibility deep-sea environments. Furthermore, Zhang *et al.*[18] focused on achieving high mechanical strength, employing a combined "superspreading and annealing" strategy to fabricate a double-network hydrogel with exceptional strength and toughness. This hydrophilic network effectively achieves the waterproofing function through external encapsulation, but it may also reduce its sensitivity and long-term stability. Additionally, Jiao *et al.*[19] constructed a dense, double-crosslinked network *via* the competitive synergistic effect between strong acids and iron ions. The resulting high crosslinking density inevitably constrains the free stretching of polymer segments, thereby reducing the material's applicability in accommodating large-deformation underwater movements. Overall, these limitations indicate that traditional designs—which rely solely on a single electrical signal and passive defense strategies like external encapsulation or dense crosslinking—cannot simultaneously address the coupled salinity, pressure, and temperature stimuli of marine environments. What is required instead is a design that moves beyond passive defense toward active environmental adaptation, integrating multimodal, synchronized responses into a single material system.

Jellyfish, ancient marine organisms, exhibit remarkable environmental adaptability by integrating transparency, luminescence, and self-healing to adjust their color, fluorescence, and movement in response to external stimuli, enabling efficient responsive navigation and self-protection in complex aquatic environments.[20–22] Inspired by the environmental adaptability and multimodal response mechanisms of jellyfish, this study envisions the development of a biomimetic smart hydrogel that combines excellent strength, multimodal sensing, and environment-responsive color-changing capabilities. Considering its targeted application in marine environments, where inevitable hydration swelling would severely compromise performance stability, the hydrophobic monomer octadecyl methacrylate (OMA) is deliberately incorporated into the system to endow the hydrogel with anti-swelling properties. Within this hydrogel matrix, the micelles formed by the cationic surfactant cetyltrimethylammonium bromide (CTAB) and the hydrophobic monomer OMA play a crucial threefold role: first, these micelles effectively solubilize and stabilize the hydrophobic segments of OMA, facilitating their micellar-confined copolymerization in the aqueous phase,[23] thereby achieving anti-swelling performance. Second, by providing an internal hydrophobic microenvironment, the micelles enable a more uniform dispersion of the color-changing microcapsules within the hydrogel network.[24] Third, the micelle-stabilized hydrophobic segments aggregate to form a hydrophobic association network; these associated domains act as high-density dynamic physical crosslinking points, significantly improving the hydrogel's overall mechanical qualities.[25]

Herein, the hydrogel is synthesized *via* one-pot micellar-confined photopolymerization, where the polymer network is copolymerized from the thermo-responsive monomer *N*-isopropylacrylamide (NIPAM), the conductive monomer 2-acrylamido-2-methylpropanesulfonic acid (AMPS), and the hydrophobic monomer OMA, alongside the synchronous incorporation of color-changing microcapsules (Fig. 1). In terms of multimodal sensing, the gel network integrates dual visual and electrical response mechanisms. On the one hand, relying on the

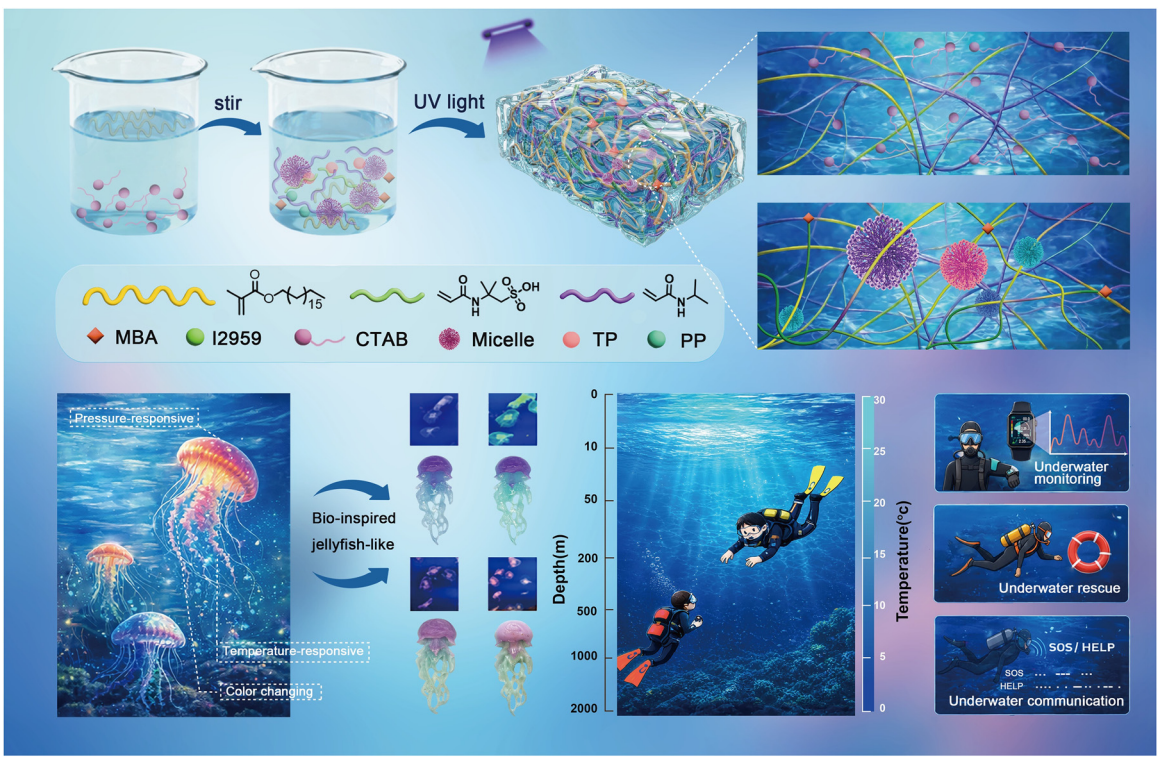


Fig. 1 A diagram of the fabrication process of PANO@C hydrogel.

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inherent temperature-sensitive phase transition characteristics of the micelles and NIPAM, combined with color-changing microcapsules, the hydrogel can mimic the environmentally adaptive color-changing behavior of jellyfish. It exhibits significant color responses under specific temperature or light stimuli, thereby enabling intuitive underwater visual warnings. On the other hand, the synergistic effect between the sulfonic acid groups in AMPS and the bromide ions released by CTAB enhances the sensing sensitivity of the hydrogel. This enables the hydrogel not only to exhibit sensitive electrical responses to tensile strain and temperature fluctuations but also to convert external pressure stimuli into real-time resistance change signals, achieving sensitive monitoring of water depth and temperature. Benefiting from its excellent anti-swelling properties, this hydrogel is highly effective for underwater motion monitoring in real time and further expanded to application scenarios such as underwater communication and emergency rescue. This work also provides new methods and perspectives for the design of intelligent wearable sensors and underwater monitoring devices.

2. Experimental section

2.1. Materials

Octadecyl methacrylate (OMA, 96%) was purchased from Macklin. *N*,*N*′-Methylenebisacrylamide (MBAA, 99%) and 2-hydroxy-4′-(2-hydroxyethoxy)-2-methylpropiophenone (I2959, 98%) were acquired from Aladdin. *N*-Isopropylacrylamide (NIPAM, 98%) and 2-acrylamido-2-methylpropanesulfonic acid (AMPS, 98%) were obtained from Macklin. Hexadecyltrimethylammonium bromide (CTAB, 99%) was acquired from Sinopharm Chemical Reagent Co., Ltd. Sodium chloride (NaCl, 99%), magnesium sulfate ($MgSO_4$, 99%), calcium chloride ($CaCl_2$, 97%), potassium chloride (KCl, 99%), and sodium bicarbonate ($NaHCO_3$, 99%) were all acquired from Macklin. Thermochromic microcapsules and photochromic microcapsules were purchased from Taobao. Throughout the experiments, deionized water was utilised.

2.2. Preparation of P(NIPAM-*co*-AMPS-*co*-OMA)@CTAB hydrogel (PANO@C)

In this study, the hydrogel was synthesized through photo-initiated radical polymerization of a micellar system consisting of NIPAM/AMPS and OMA/CTAB. Initially, CTAB was dissolved in deionized water under stirring at 50 °C until completely dissolved. Next, OMA was incorporated into the solution, followed by stirring for an additional 90 min to form a clear and stable micellar solution. Subsequently, NIPAM, AMPS, the photoinitiator I2959, the crosslinker MBAA and color-changing microcapsules were included sequentially, and the mixture was agitated at 30 °C for 60 min to achieve a uniform precursor solution. The precursor solution was then subjected to ultrasonication for 5 min to eliminate bubbles and poured into a PTFE mold. Polymerization was carried out under 365 nm UV light (intensity: 50 mW cm^{-2}) for 15 min, resulting in the formation of the PANO@C hydrogel. To investigate the effect of different amounts of the hydrophobic monomer OMA on the hydrogel properties, six different variants of the PANO@C hydrogel were prepared. The formulation details for each sample are outlined in Table S1, with the samples labeled as $PANO_x$@C, where "x" stands for the mass proportion of OMA within the hydrogel. In all formulations, the mass ratio of CTAB to OMA was set at 5 : 1.

2.3. Preparation of artificial seawater

At 25 °C, NaCl (26.726 g), $MgSO_4$ (3.248 g), $CaCl_2$ (1.153 g), KCl (0.721 g), $NaHCO_3$ (0.198 g), and deionized water (1000 mL) were mixed together and stirred for 2 hours to obtain artificial seawater.

2.4. Characterization

The PANO@C hydrogel's cross-sectional shape was analyzed using scanning electron microscopy (SEM, Regulus 8100), whereas the elemental composition of the surface was determined using energy-dispersive X-ray spectroscopy (EDS). The chemical composition of the hydrogel was characterized by Fourier-transform infrared spectroscopy (FT-IR, Nicolet iS50, Thermo Fisher Scientific, USA) and X-ray photoelectron spectroscopy (XPS, ESCALAB Xi+, Thermo Fisher Scientific, USA). Thermal properties were assessed using thermogravimetric analysis (TGA, Q500, USA) and differential scanning calorimetry (DSC, Q2000). Dynamic light scattering (DLS, Zetasizer Ultra) was used to assess the particle size distribution. Sensor performances related to strain, pressure, and temperature were measured using an electrochemical workstation (CHI 760E). Contact angle measurements were performed using a DAS100 goniometer.

Mechanical performance of the hydrogel was evaluated through tensile, adhesion, and compression tests. Tensile tests utilized dumbbell-shaped specimens (dimensions: 20 × 4 × 2 mm) on a QJ211S-5 kN universal testing machine at a speed of 100 mm min^{-1}, with each test repeated three times for consistency. Adhesion performance of the hydrogel (sample size: 10 × 8 × 1 mm) was tested using the same device at a stretching speed of 30 mm min^{-1}, also with three repetitions. Compression tests were conducted on cylindrical samples (diameter: 26 mm, height: 10 mm) using an LE3504 universal testing machine at a compression rate of 6 mm min^{-1}, with three repetitions to ensure reliability. Electrical conductivity, along with the application of the hydrogel in human motion monitoring, was examined with an electrochemical workstation CHI750E. The gauge factor (GF) of the hydrogel was determined according to the following equation:

$$\mathrm{GF} = \frac{\Delta R}{R_0 \times \Delta\varepsilon} \quad (1)$$

where ΔR (defined as $R - R_0$) is the resistance change, $\Delta\varepsilon$ is the applied strain, and R_0 is the initial resistance.

The antibacterial performance of PANO@C was tested using the shake flask method. *Escherichia coli* (*E. coli*) and *Staphylococcus aureus* (*S. aureus*) were selected as test strains. Three groups were established: a blank control group, a PNA hydrogel group, and a PANO@C hydrogel group. Each group was incubated with the bacterial suspension at 37 °C under shaking conditions, with three parallel samples tested for each group.

The antibacterial efficacy of PANO@C was determined by calculating the antibacterial rate.

2.5. Ethical statement

The study was carried out in compliance with applicable ethical norms and the Declaration of Helsinki's tenets. Every individual willingly participated in this study and gave their informed consent. The Jiangnan University Medical Ethics Committee has approved the experiment in its entirety (ID: JNU202601RB004).

3. Results and discussion

3.1. Characterization of PANO@C hydrogel

The composition and structure of the hydrogels were determined using FTIR analysis. As illustrated in Fig. 2a, the broad peak of PANO@C around 3351 cm^{-1} is attributed to the structured water's O–H stretching vibration, along with the N–H stretching vibration of the amide bonds from the NIPAM units.[26,27] The distinctive isopropyl C–H bond deformation vibrations on the PNIPAM chains can be found at 1390 cm^{-1} and 1367 cm^{-1}.[28] A peak at 1247 cm^{-1} is ascribed to the S═O stretching vibration of sulfonic acid groups in the AMPS segments. OMA shows a C═O stretching vibration signal at 1720 ?A3B2 tlsb -0.045w?>cm^{-1},[29] and absorption peaks near 2920 cm^{-1} and 2850 cm^{-1} are associated with the stretching vibrations of $–CH_2–$ and $–CH_3$, confirming the existence of long alkyl chains in the polymer. The intensity of these peaks increases with higher OMA content (Fig. 2b), suggesting that the incorporation of these hydrophobic structural units promotes the establishment of hydrophobic microdomains within the network. This structural enhancement contributes to improved mechanical properties and anti-swelling behavior. Additionally, the disappearance of the C═C stretching vibration peak indicates the successful synthesis of the PANO@C hydrogel. As shown in Fig. 2c, the PANO@C hydrogel displays two broad diffraction peaks at approximately 9.6° and 20.3°, confirming its amorphous nature.[30] As the OMA content increased, the broad diffraction peak near $2\theta = 9.6°$ gradually shifted to lower angles, indicating an increased *d*-spacing associated with the expansion of hydrophobic association domains or micellar aggregates. In contrast, the broad peak near $2\theta = 20.3°$ shifted to higher angles, suggesting decreased interchain spacing and thus denser local packing of polymer segments.[31]

The LCST and UCST of hydrogels with varying OMA contents were further investigated. As shown in Fig. 2d, increasing OMA content decreases the LCST, due to the enhanced hydrophobicity and weakened hydrogen bonding, which shifts the hydrophilic–hydrophobic balance toward hydrophobic interactions at lower temperatures. Simultaneously, the combined increase in OMA and CTAB contents strengthens hydrophobic interactions and micellar structures, promoting polymer chain association and stabilizing the phase-separated state at lower temperatures. As a result, higher thermal energy is needed to disrupt these interactions, leading to an increase in UCST. Moreover, Fig. 2e shows that all PANO@C hydrogels display similar thermal degradation profiles, demonstrating good thermal stability. In the main decomposition stage (200–400 °C),[32] the residual carbon content remains above 37%. Specifically, $PANO_{2.5}$@C exhibits a maximum degradation rate around 323 °C, followed by a second degradation stage near 424 °C, corresponding to further decomposition and carbonization of the polymer backbone (Fig. 2f).

To verify that the hydrogel contains micelles, DLS measurements were performed. A scattering peak at approximately 1.71 μm in the prepolymer solution indicated the formation of micelles (Fig. 2g). These micelles serve as crosslinking sites, enhancing the crosslinking density of the polymer network and stabilizing polymer network structure.[33,34] The zeta potentials of both samples exceeded 30 mV (Fig. 2h), suggesting strong surface charges on the particles. The electrostatic repulsion effectively prevents aggregation and enhances the dispersion stability of the system.[35,36] Notably, $PANO_{2.5}$@C exhibited a higher zeta potential, suggesting superior colloidal stability. The PANO@C hydrogel exhibits dual thermoresponsive behavior. As shown in Fig. 2i, when the temperature decreases, OMA aggregates within CTAB micelles, causing the micelle size to increase[37] and the hydrogel's transparency to decrease (transmittance: 41.8%). Upon heating above the UCST, the micelle size decreases, crystals begin to dissolve, and the hydrogel becomes transparent again (transmittance: 91.5%). When the temperature surpasses the LCST, the polymer chains undergo phase separation, exposing hydrophobic groups and causing network contraction and pore size reduction. This results in enhanced light scattering and a sharp drop in transparency (transmittance: 19.3%).

Fig. 2j shows the planar morphology of the PAN hydrogel, revealing a highly porous structure. In contrast, after introducing the hydrophobic monomer OMA, the pore size of the $PANO_{2.5}$@C hydrogel remarkably decreased, and the overall structure became denser (Fig. 2k–m and Fig. S1), corresponding to its enhanced anti-swelling performance. The elemental distribution of the $PANO_{2.5}$@C hydrogel is shown in Fig. 2n and o, where the hydrogel is composed of C, N, O, S, and Br, with all elements distributed with uniformity. XPS analysis was also conducted for hydrogel. The C 1s spectrum can be broken down into three characteristic peaks (Fig. S2a),[38] corresponding to C–C/C–H (284.4 eV), C–O/C–N (285.8 eV), and C═O (288.3 eV). Fig. S2b shows the O 1s spectra having peaks at 531.3 eV and 532.8 eV, assigned to C–O–C and C═O bonds,[39] respectively. The N 1s spectrum exhibits peaks at 401.5 eV and 402.1 eV (Fig. S2c), corresponding to N–H and C–N bonds,[40] respectively. The XPS results further validate the successful formation of the hydrogel matrix and verify its expected surface chemical composition.

3.2. Mechanical properties of PANO@C hydrogel

For wearable sensors, excellent mechanical properties are essential for practical applications. Therefore, the hydrogel material must possess good tensile and compressive strength, along with exceptional stability. As shown in Fig. 3a, the hydrogel demonstrates excellent tensile performance, with a strain exceeding 15 times during stretching. It also maintains structural integrity under complex deformations, such as knotting, twisting, and bending, reflecting its good flexibility. Moreover, the hydrogel can rapidly revert to its original shape after compression and is

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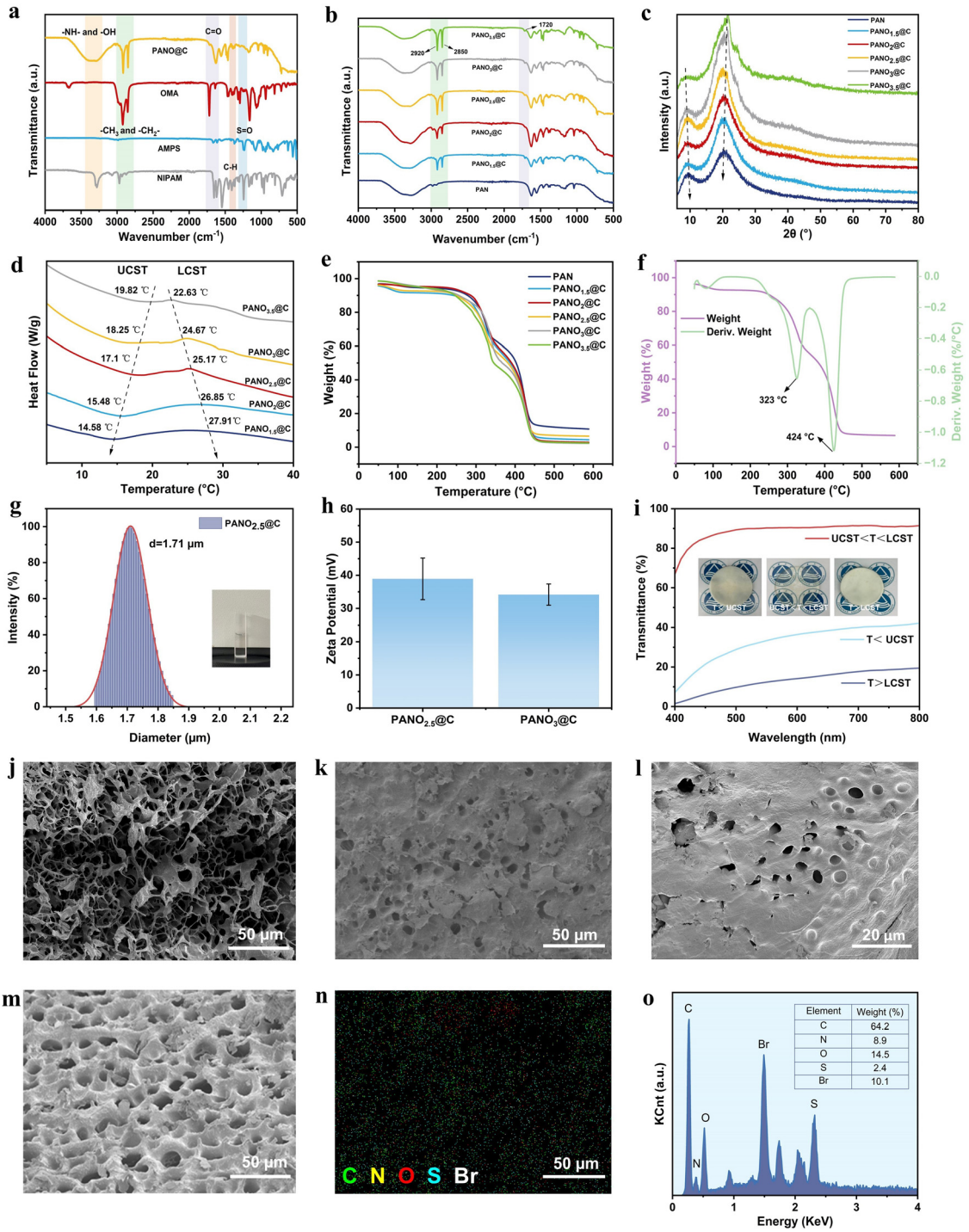


Fig. 2 (a) FTIR spectra of NIPAM, AMPS, OMA and PANO@C; (b) FTIR spectrum of PANO@C hydrogel; (c) XRD patterns of PANO@C hydrogel; (d) DSC curves of PANO@C hydrogel; (e) TG curves of PANO@C hydrogel; (f) TG and DTG curves of $PANO_{2.5}$@C hydrogel; (g) DLS results of $PANO_{2.5}$@C precursor solutions; (h) Zeta potentials of $PANO_{2.5}$@C and $PANO_3$@C precursor solutions; (i) Transmittance spectra of $PANO_{2.5}$@C hydrogel at different temperatures; (j) SEM image of the surface of PAN hydrogel; (k–l) SEM images of the surface of $PANO_{2.5}$@C hydrogel; (m) SEM image of the cross-section of $PANO_{2.5}$@C hydrogel; (n) and (o) EDS spectra and elemental analysis of $PANO_{2.5}$@C hydrogel.

capable of lifting a heavy object of approximately 2.5 kg (Fig. 3b), further confirming its superior mechanical strength. The OMA content affects the PANO@C hydrogel's mechanical characteristics. As shown in Fig. 3c and d, increasing OMA content leads to an increase in fracture strength tends to increase, while the elongation at break first increases and then decreases. This behavior can be attributed to the formation of more hydrophobic interactions within the network due to the introduction of OMA. Meanwhile, under the influence of CTAB, more physical cross-linking points and electrostatic interactions are formed, resulting

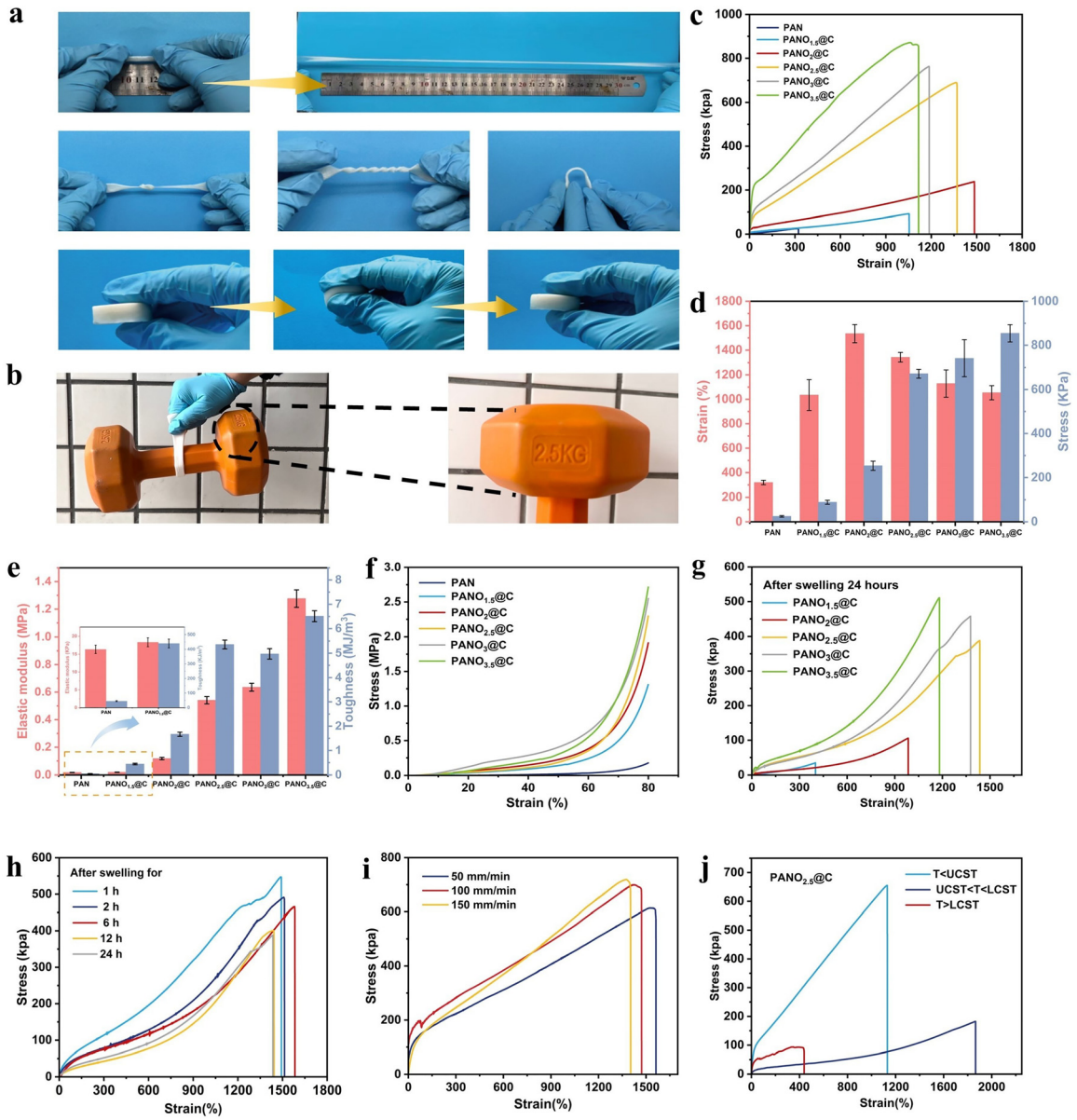


Fig. 3 Photographs of mechanical properties for PANO@C hydrogels: (a) demonstration of stretching, knotting, twisting, bending, and compression; (b) load-bearing capability; (c) stress–strain curves of various PANO@C hydrogels; (d) fracture strength and elongation at break; (e) toughness and elastic modulus; (f) compressive stress–strain curves for different PANO@C hydrogels under 80% compression; (g) stress–strain curves after 24 h of swelling; (h) stress–strain curves after swelling for different durations; (i) stress–strain curves at different stretching rates; (j) stress–strain curves under different temperatures.

in a denser network structure and significantly enhanced mechanical strength. However, when the OMA and CTAB contents further increased, the excessively high crosslinking density limits the hydrogel's deformability, making it more brittle and rigid.[33] Among the different formulations, $PANO_{2.5}$@C shows an elongation at break of 1370% and a fracture strength of 690 kPa, representing a significant improvement over $PANO_2$@C. When the OMA content is increased further to 3% and 3.5%, the enhancement in mechanical properties becomes less pronounced. Based on the overall trend of mechanical performance, $PANO_{2.5}$@C was selected as the optimal formulation for later experimental procedures. Additionally, the elastic modulus of the hydrogel increases gradually from 16.3 kPa for PAN to 1273 kPa for $PANO_{3.5}$@C (Fig. 3e), indicating that the hydrogel becomes more resistant to deformation as the OMA and CTAB contents increase. The toughness also rises from 0.045 MJ m^{-3} for PAN to 6.48 MJ m^{-3} for $PANO_{3.5}$@C. In terms of compressive performance, the hydrogel can withstand higher stress under the same strain as the OMA content increases, indicating improved compressive strength (Fig. 3f). Since the hydrogel is designed for underwater sensing, it is essential that it retains excellent mechanical properties and fatigue resistance after swelling. The PANO@C hydrogel was submerged in deionized water for 24 hours to achieve swelling equilibrium, after which its tensile

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properties were tested. As shown in Fig. 3g, the swollen $PANO_{2.5}$@C hydrogel still exhibits excellent elongation at break (1435%) and fracture strength (387 kPa).

To further elucidate this mechanical evolution, the tensile properties were monitored dynamically over the immersion period (Fig. 3h). With the extension of swelling time, the fracture strength gradually decreased, while the elongation at break remained largely unchanged. This initial decrease in strength can be attributed to the water-induced plasticization effect: the infiltrated water molecules expanded the free volume of the polymer network and weakened a portion of the rigid intermolecular hydrogen bonds, thereby reducing the network's load-bearing capacity. Meanwhile, acting as an internal lubricant, the water molecules promoted the sliding and disentanglement of polymer chains during stretching, enabling the network to maintain high extensibility despite the drop in strength. Notably, the fracture strength tended to stabilize after 12 h of immersion, demonstrating that the hydrogel had reached swelling equilibrium and the network structure would not undergo continuous degradation. The hydrogel's anti-swelling qualities were also analyzed. After long-term immersion in water, $PANO_{2.5}$@C hydrogel maintained a stable morphology (Fig. S3a). The water contact angle was found to be approximately 115°, and remained above 110° after 15 s (Fig. S3b), indicating a moderate level of hydrophobicity. The swelling ratio gradually increased and stabilized at approximately 31.2% (Fig. S3c), confirming good anti-swelling performance. Accordingly, the $PANO_{2.5}$@C hydrogel combines excellent mechanical properties with strong anti-swelling stability, making it a suitable material for underwater sensing applications.

The impact of varying stretching rates on the mechanical properties of the hydrogel was further analyzed. As shown in Fig. 3i, as the stretching rate elevated from 50 mm min^{-1} to 150 mm min^{-1}, the stress at the same strain gradually increased, while the elongation at break slightly decreased. This behavior is attributed to the fact that under high-speed stretching, the polymer chains do not have sufficient time to fully extend and rearrange, resulting in a more rigid response. Meanwhile, the rapid deformation restricts chain mobility, leading to earlier fracture. It is worth noting that the aforementioned standard mechanical tests (including Fig. 3c and i) were conducted when the ambient temperature was naturally below the UCST. At this temperature, the local freezing of the OMA segments and the restricted chain motion endow the hydrogel with an optimally toughened state. As shown in Fig. 3j, temperature serves a vital part in the mechanical properties of the $PANO_{2.5}$@C hydrogel. When the temperature is below the upper critical solution temperature (UCST), the hydrophobic interactions among CTAB micelles are strengthened. The low temperature also restricts the movement of both molecular chain segments and micelles, resulting in a more stable micellar structure and a greater number stable physical crosslinking points within the network. Consequently, the hydrogel exhibits a relatively high fracture strength (approximately 650 kPa) while maintaining a high elongation at break (approximately 1130%). When the temperature rises above UCST but remains below the lower critical solution temperature (LCST), the elongation at break increases to approximately 1850%, while the strength decreases to around 180 kPa. This indicates that as the temperature rises, some physical crosslinking points are weakened, leading to a softer network structure and increased chain segment mobility.[41] Similarly, the compressive properties in this temperature range follow a similar trend, with a noticeable decrease in compressive stress compared to that at temperatures below the UCST, which is in line with the observed reduction in tensile strength (Fig. S4). When the temperature exceeds the LCST, the hydrophobic collapse of the NIPAM chains further reduces both the elongation at break and the fracture strength of the hydrogel, bringing them down to approximately 420% and 90 kPa, respectively. Fig. S5a presents the loading–unloading curves of the hydrogel under 100% strain over 10 cycles. The hydrogel exhibits a slight hysteresis during the first deformation–recovery cycle, which is significantly improved during subsequent 2nd to 10th cycles. This behavior is attributed to the synergistic effect of multiple hydrogen bonds and electrostatic interactions within the hydrogel. The first loading–unloading cycle exhibits an energy dissipation of 33.27 kJ m^{-3}, decreasing to 17.93 kJ m^{-3} in the second cycle and gradually declining in subsequent cycles before stabilizing at 12.26 kJ m^{-3} (Fig. S5b). These results demonstrate that the $PANO_{2.5}$@C hydrogel possesses outstanding energy dissipation capability and fatigue resistance under repeated compression loading–unloading cycles.

3.3. Adhesion and antibacterial properties of PANO@C hydrogel

As shown in Fig. 4a, the PANO@C hydrogel can firmly adhere to a range of surfaces such as rubber, glass, plastic, fabric, wood, and metal. The lap shear test was employed to evaluate the adhesion performance of the hydrogel (Fig. S6) revealing a maximum adhesion strength of 60 kPa (Fig. 4b). The adhesive performance of the hydrogel results from the combined effects of its components: the thermoresponsive NIPAM enhances adhesion above the LCST, while the sulfonic acid groups of AMPS offer multiple non-covalent interactions. Notably, after five cycles of use, the hydrogel maintains its adhesion with only a slight reduction in strength (Fig. 4c). Moreover, the hydrogel can be easily removed from the skin without leaving residue or causing irritation (Fig. S7).

Ensuring that surfaces remain free of bacterial growth is vital for the functionality of wearable sensors and to prevent infection. In this research, *Escherichia coli* and *Staphylococcus aureus* were utilised as model organisms to evaluate antibacterial effectiveness. PNA and PANO@C were tested for their antibacterial properties (Fig. 4e and f). PNA exhibits antibacterial rates of 96.15% against *E. coli* and 94.48% against *S. aureus*, due to its sulfonic acid groups, which ionize strongly in water, releasing hydronium ions that lower the pH and apply stress to bacterial membranes, causing their rupture.[42,43] PANO@C exhibited enhanced antibacterial rates of 99.23% against *E. coli* and 98.7% against *S. aureus*, thanks to a combination of sulfonic acid groups, positively charged quaternary ammonium groups, and hydrophobic alkyl chains.[44] The sulfonic

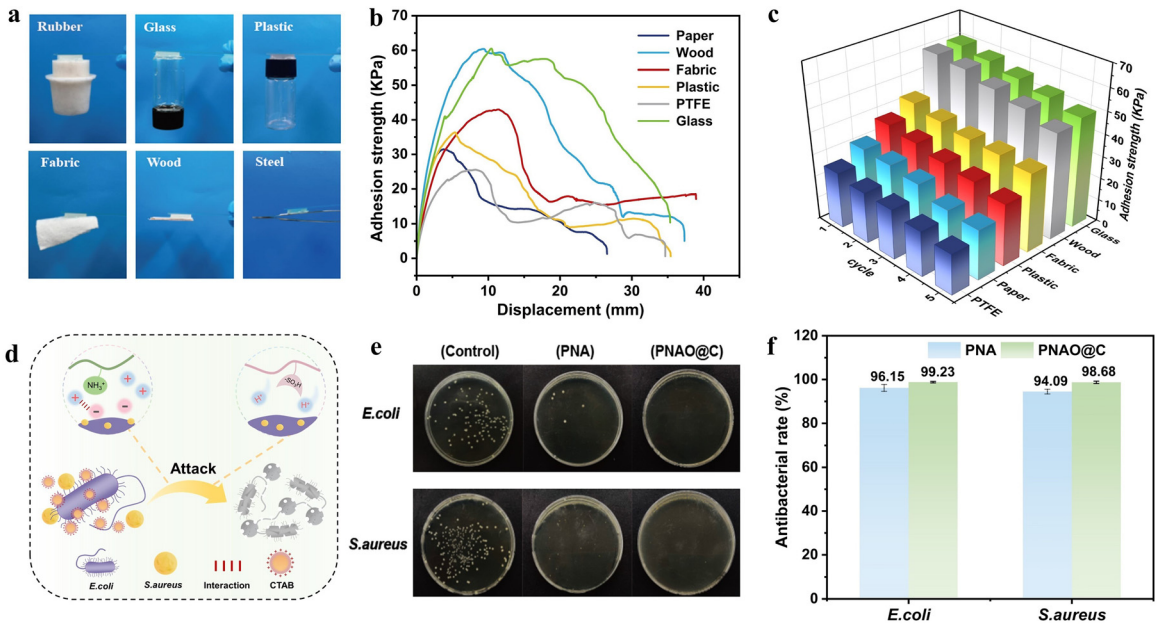


Fig. 4 (a) Photographs showing the hydrogel firmly adhering to various materials, including rubber, glass, plastic, fabric, wood, and metal. (b) Peel curves of the hydrogel on various substrates. (c) Five repeated adhesion cycles of the hydrogel on different substrates. (d) An illustration of the antibacterial mechanism. (e) Antibacterial effects of PNA and PNAO@C hydrogels against *E. coli* and *S. aureus*. (f) Comparison of antibacterial rates of PNA and PNA@O hydrogels against *E. coli* and *S. aureus*.

acid groups acidify and disrupt membrane integrity, quaternary ammonium groups attach electrostatically to the negatively charged bacterial surface, and hydrophobic alkyl chains penetrate the phospholipid bilayer, further compromising the membrane structure (Fig. 4d).

3.4. Stress sensing performance of PANO@C hydrogel

The brightness of the LED in the conductive circuit gradually decreases with increasing applied strain to the hydrogel (Fig. S8a). This behavior can be attributed to the elongation of the internal conductive pathways during stretching, which increases the resistance, thereby reducing the current and causing the LED to dim. Remarkably, even at a high elongation of 1200%, the hydrogel still maintains favorable electrical conductivity, indicating a stable electrical response under large deformations and demonstrating its suitability for large-strain sensing applications. Moreover, the hydrogel is capable of clearly distinguishing mechanical deformations of different magnitudes. As shown in Fig. 5a–c, stable and repeatable signal outputs are observed throughout the strain ranges of small (1–5%), medium (10–50%), and large (100–300%), indicating its strong capability to monitor both subtle physiological motions and large-scale movements. This highlights its promising potential for wearable sensing applications. The sensitivity of the hydrogel sensor, expressed as the gauge factor (GF), is a key performance parameter described as the relative resistance change ratio to applied strain. As illustrated in Fig. 5d, the GF increases with strain. Specifically, three distinct regions can be identified: 1.53 in the 0–200% strain range, 2.35 in the 200–400% range, and 2.58 in the 400–700% range. As illustrated in Fig. 5e, under a strain of 3%, the $PANO_{2.5}$@C hydrogel exhibits a rapid response, with response and recovery times of 204 ms and 218 ms, correspondingly. Fig. 5f shows the relative resistance changes under different stretching frequencies (50, 100, 150, 200, and 250 mm min^{-1}). The results demonstrate that the electrical response curves remain highly consistent across varying deformation rates, suggesting that frequency has a negligible effect on sensing stability. This further confirms that the hydrogel can deliver reliable and reproducible electrical signals under identical strain conditions. For practical applications, signal stability and durability are also crucial factors in evaluating sensing performance. The hydrogel sensor was repetitively loaded–unloaded for 500 cycles at 50% strain (Fig. S8b). The relative resistance response remains largely stable throughout the cycles, demonstrating excellent fatigue resistance and long-term operational stability.

To simulate the effects of varying underwater depths on the sensor, uniaxial compression tests were conducted to apply mechanical pressure to the hydrogel. According to the hydrostatic pressure equation ($P = \rho gh$), where depth (h) is linearly related to hydrostatic pressure (P), mechanical pressure in air was used to an analog for water pressure, with a conversion factor of approximately 10 kPa per 1 meter of water depth. As shown in Fig. 5g, increasing compression—equivalent to water pressure—results in greater the brightness of the LED in the conductive circuit gradually increases. This is due to the compression of the hydrogel's internal microstructural network, forming denser conductive pathways that reduce resistance and increase current, thereby enhancing LED brightness. Fig. 5h indicates that under 3 kPa of compression (equivalent to 0.3 meters of a water depth), the $PANO_{2.5}$@C hydrogel responds quickly, with response and recovery times of 220 ms and 357 ms, respectively. The gauge factor (GF) illustrated in Fig. 5i, can be

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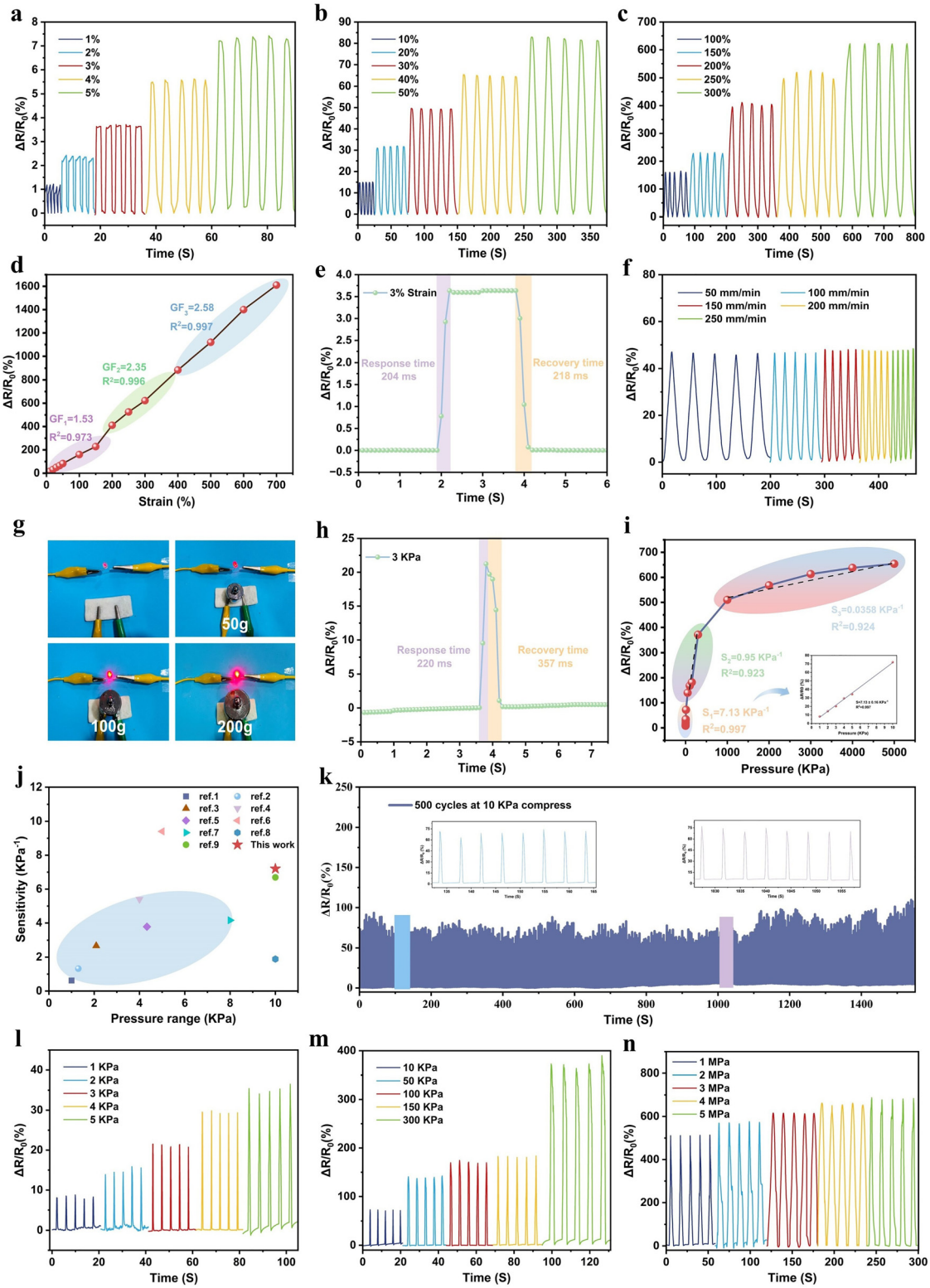


Fig. 5 (a–c) Resistance change curves of $PANO_{2.5}$@C hydrogel under different strain conditions. (d) Relative resistance–strain response curve of $PANO_{2.5}$@C hydrogel. (e) Response time and recovery time of $PANO_{2.5}$@C hydrogel under stretching. (f) Resistance curves of $PANO_{2.5}$@C hydrogel under stretching at different frequencies. (g) Variation of LED brightness under compression of the hydrogel; (h) response and recovery times of $PANO_{2.5}$@C hydrogel under compression; (i) relative resistance–pressure response curve of $PANO_{2.5}$@C hydrogel; (j) comparison of GF of $PANO_{2.5}$@C hydrogel with results from in other studies; (k) relative resistance change of $PANO_{2.5}$@C hydrogel after 500 compression cycles at 10 kPa; (l–n) resistance change curves of $PANO_{2.5}$@C hydrogel under different compressions conditions.

categorized into three regions: 7.13 kPa^{-1} for 0–10 kPa (0–1 m depth), 0.95 kPa^{-1} for 10–300 kPa (1–30 m depth), and 0.0358 kPa^{-1} for 300–5000 kPa (30–500 m depth). The high sensitivity of the hydrogel is mainly attributed to its porous structure. Compression brings pore walls into contact, significantly increasing the effective interfaces for charge transfer and polarization, allowing small deformations to induce considerable charge accumulation.[45]

Compared with other hydrogel-based pressure sensors (Fig. 5j), the $PANO_{2.5}$@C hydrogel exhibits excellent sensitivity.[46–54] As revealed in Fig. 5k, the sensor maintains stable electrochemical signal output even after 500 compression cycles at 10 kPa and under various compression rates (30–150 mm min^{-1}) (Fig. S9). Fig. 5l and m further show that the relative resistance change of the $PANO_{2.5}$@C hydrogel increases with rising compression from 1 kPa to 300 kPa. Accordingly, the $PANO_{2.5}$@C hydrogel exhibits sensitive, stable, and reproducible pressure responses over a wide substantial range (0–5000 kPa, corresponding to a water depth of 0–500 meters), demonstrating its potential as an effective underwater pressure sensor for depth monitoring.

3.5. Temperature sensing performance of PANO@C hydrogel

In the marine environment, water temperature decreases significantly as depth increases, showing a clear negative correlation between depth and temperature (Fig. 6a). This relationship allows the temperature-responsive properties of hydrogel to serve as an indirect indicator of underwater depth. Specifically, as shown in Fig. 6b, when the hydrogel interacts with a cold source (simulating conditions in deep-water environments), the low temperature environment impedes ion migration, resulting in an increase in system resistance, a decrease in current, and a corresponding weakening of LED brightness. Conversely, exposure to a heat source (simulating shallow-water conditions) triggers hydrophobic collapse in NIPAM chains, causing network densification. The elevated temperature accelerates polymer segment dynamics and promotes the thermal dissociation of tight ion pairs, providing abundant transient hopping sites and free charge carriers. Consequently, this transformation enhances the continuity of ion transport pathway, increases ion migration rates, reduces resistance, raises current, and brightens the LED. Fig. 6c demonstrates that the hydrogel maintains stable sensing performance under

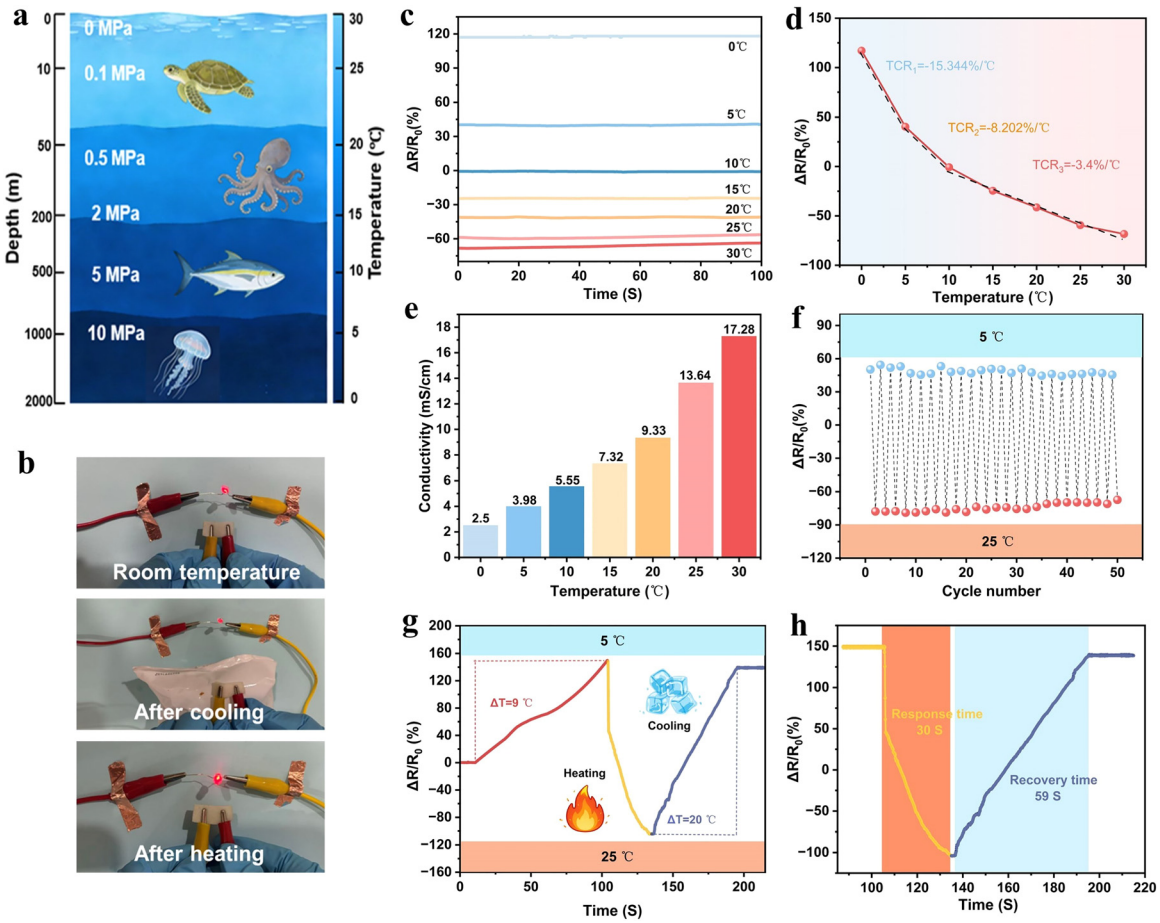


Fig. 6 (a) Ocean depth and corresponding pressure at various temperature; (b) variation of LED brightness with temperature of the hydrogel; (c) stability of $PANO_{2.5}$@C hydrogel at different temperatures; (d) temperature sensitivity of $PANO_{2.5}$@C hydrogel; (e) electrical conductivity of $PANO_{2.5}$@C hydrogel at different temperatures; (f) reproducible temperature temperature-sensing performance of $PANO_{2.5}$@C hydrogel under repeated exposure to a cold source (5 °C) and a heat source (25 °C); (g) response curves of $PANO_{2.5}$@C hydrogel at varying temperatures; (h) corresponding response and recovery times.

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different water temperatures (0, 5, 10, 15, 20, 25, and 30 °C), with negligible signal drift. These temperatures correspond to distinct ocean depths: tropical surface seawater (0–10 m) aligns with 30 °C, the upper thermocline (~50–100 m) corresponds to 20 °C, the middle thermocline (~200–500 m) relates to 10 °C, and deep seawater (>1000 m) aligns with 0–5 °C. The results confirm the hydrogel's reliable and effective depth-sensing capabilities. Fig. 6d presents the temperature sensitivity curve of the hydrogel, showing its highest sensitivity in the 0–5 °C range (−15.344%/°C), followed by −8.202%/°C between 5–10 °C and −3.4%/°C form 10–30 °C. Additionally, as the temperature rises from 0 °C to 30 °C, the electrical conductivity of the hydrogel displays a steady monotonic rise, expanding from approximately 2.5 mS cm^{-1} to 17.28 mS cm^{-1} (Fig. 6e). This result indicates that the system possesses significant temperature-responsive conductive properties. At lower temperatures, restricted polymer chain movement hinders ion migration, but as temperatures rise, chain relaxation facilitates ion transport, boosting conductivity. In the 20–30 °C range, conductivity increases more rapidly, likely due to conformational changes in NIPAM chains. The hydrogel also exhibits impressive cycling stability, as confirmed by Fig. 6f. Following 50 heating/cooling cycles at 5 °C and 25 °C, the resistance response remains consistent. Dynamic testing (Fig. 6g) reveals that the hydrogel produces distinct, reproducible electrical signal

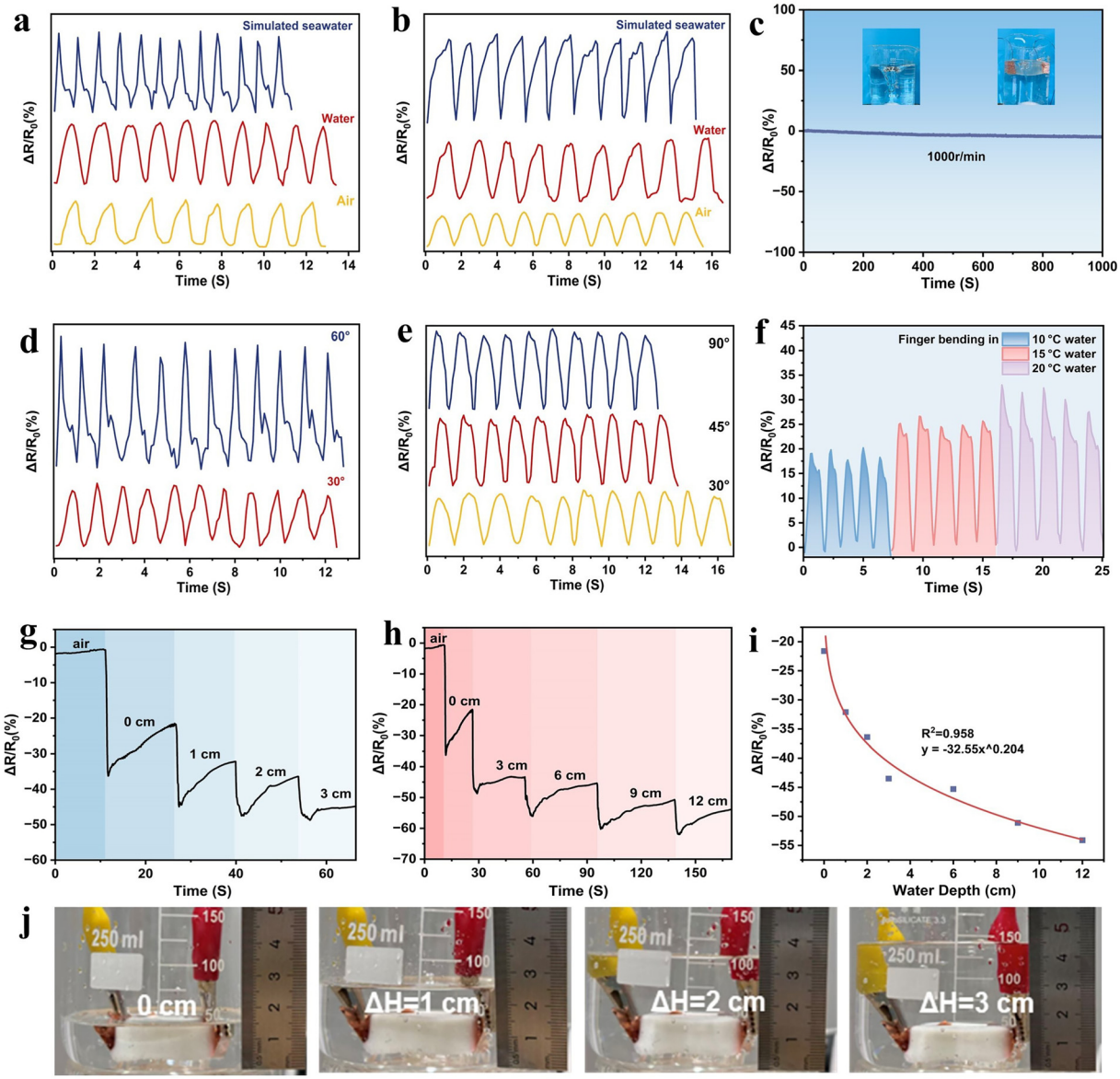


Fig. 7 (a) Resistance change curves of wrist movement in air, water, and seawater; (b) resistance change curves of finger movement in air, water, and seawater; (c) stability of the hydrogel under vortex conditions; (d) resistance change curves of wrist movement at different angles in seawater; (e) resistance change curves of finger movement at different angles in seawater; (f) relative resistance change of finger bending at different temperatures; (g) and (h) resistance changes of $PANO_{2.5}$@C hydrogel at different water depths; (i) fitting relationship between resistance response and water depth; (j) photograph of the hydrogel used for water level detection.

responses to external temperature fluctuations. Furthermore, Fig. 6h highlights its response and recovery times of 30 s and 59 s, respectively, reflecting excellent response speed and reversibility.

3.6. Underwater stress sensing performance of PANO@C hydrogel

To assess the practical application potential of the sensor in underwater environments, its sensing performance was tested under various different media and dynamic conditions. As shown in Fig. 7a and b, the sensor was attached to a human wrist and finger, respectively, and resistance changes during repetitive movements were recorded in air, pure water, and simulated seawater the results demonstrated that the sensor produced stable and periodic electrical signals across all tested media. Notably, the hydrogel exhibited the highest resistance change in simulated seawater, followed by pure water, with air yielding the smallest response. In simulated seawater, the influx of external electrolyte ions into the hydrogel network significantly enhanced initial conductivity, amplifying the effects of deformation-induced ion channel changes, thereby generating the largest response. In contrast, pure water, lacking external ion compensation, produced a moderate response. In air, where only the internal conductive network responded to mechanical deformation, the signal was relatively minimal. These results demonstrate that the sensor has higher sensitivity in ion-rich underwater environments such as marine monitoring. Further simulations of complex underwater conditions, including strong vortex environments, are presented in Fig. 7c. Under rotational interference at 1000 rpm, the hydrogel sensor displayed stable resistance signals without significant fluctuations. The response of the sensor to different wrist and finger bending angles was also evaluated in a seawater medium. As shown in Fig. 7d and e, the relative variation in resistance consistently increased with greater bending angles.

In addition, the influence of dynamic temperature variations in underwater environments was studied. Fig. 7f compares the relative variation in resistance of the hydrogel during finger bending under diverse water temperatures. Results revealed that the resistance response gradually increased as water temperature rose. This phenomenon is attributed to the temperature-responsive conformational changes of the PNIPAM chains below the LCST, and the disassembly of micelles approaching their UCST. At lower temperatures (*e.g.*, 10 °C), micelles are aggregated, restricting ion migration. As the temperature increases to 20 °C, chain segment mobility and ion migration are enhanced, with partial chain segments gradually approaching phase transition states that adjust the network microstructure, thereby improving ion migration and amplifing the resistance response.

To further assess the multi-parameter response capability of the hydrogel in practical underwater environments, its resistance changes were tested under different water depths. A simulated experiment involved incrementally adding water to a beaker while recording real-time resistance changes.[55] As shown in Fig. 7g and h. The relative resistance change ($\Delta R/R_0$) rapidly decreased upon water addition and stabilized when water addition ceased. With the gradual rise in water level, the sensor showed a clear stepwise negative correlation between water depth and resistance change. Fitting the experimental data yielded a high correlation coefficient (R^2 = 0.958), verifying the linear relationship between water depth and sensor response (Fig. 7i). The water addition process is illustrated in Fig. 7j. These results demonstrate the hydrogel sensor's robust sensitivity and stable performance across variable underwater conditions, including media composition, temperature, depth, and dynamic interference, showcasing its potential for diverse underwater sensing applications.

3.7. Detection and sensing behavior of PANO@C hydrogel sensor

Considering the excellent strain sensing performance of the $PANO_{2.5}$@C hydrogel, a wearable flexible strain sensor was developed to monitor both large-scale and subtle human body movements. As shown in Fig. 8a, the hydrogel sensor was affixed to various parts of the human body to track different body movements. When placed on the finger, the sensor accurately detected changes in fingertip bending at angles of 30°, 45°, and 90°, with the relative resistance change increasing proportionally with the bending angle (Fig. 8b). Beyond finger movements, the sensor consistently monitored joint motions of the elbow, ankle, wrist, and knee (Fig. 8c–f), demonstrating fast response times and reliable repeatability during flexion and extension. Importantly, the hydrogel sensor is capable of capturing both large-scale body movements, such as kicking and paddling, and subtle mechanical signals like minor muscle exertions (Fig. 8g). This sensitivity is particularly valuable for evaluating muscle fatigue or identifying abnormal movement patterns. In addition, the sensor effectively distinguished finer actions, including fist clenching and finger pressing (Fig. 8h–i), highlighting its versatility. Accordingly, these results demonstrate that the $PANO_{2.5}$@C hydrogel strain sensor holds significant promise for human motion monitoring, providing a feasibility basis for subsequent underwater applications such as diver performance assessment and safety warning systems.

3.8. Underwater communication and rescue monitoring application of PANO@C hydrogel

To further validate the practical potential of the $PANO_{2.5}$@C hydrogel sensor, its performance was tested in simulated scenarios involving underwater communication and emergency rescue monitoring (Fig. 9a). Drawing inspiration from the bioluminescent and color-changing capabilities of jellyfish, thermosensitive and photosensitive functional components were integrated into the hydrogel matrix, creating a biomimetic multidimensional response system that displayed dynamic color variations under light and temperature stimuli (Fig. 9b). In a simulated rescue situation, the hydrogel sensor fulfilled dual functions of visual warning and electrical monitoring. On one hand, leveraging the temperature-dependent color-changing characteristics of the hydrogel, divers can visually assess the condition of a trapped companion through direct observation or under ultraviolet light. On the other hand, the sensor transmits real-time resistance signals generated by limb movements under varying temperatures conditions (correlating with different water depths) to a remote

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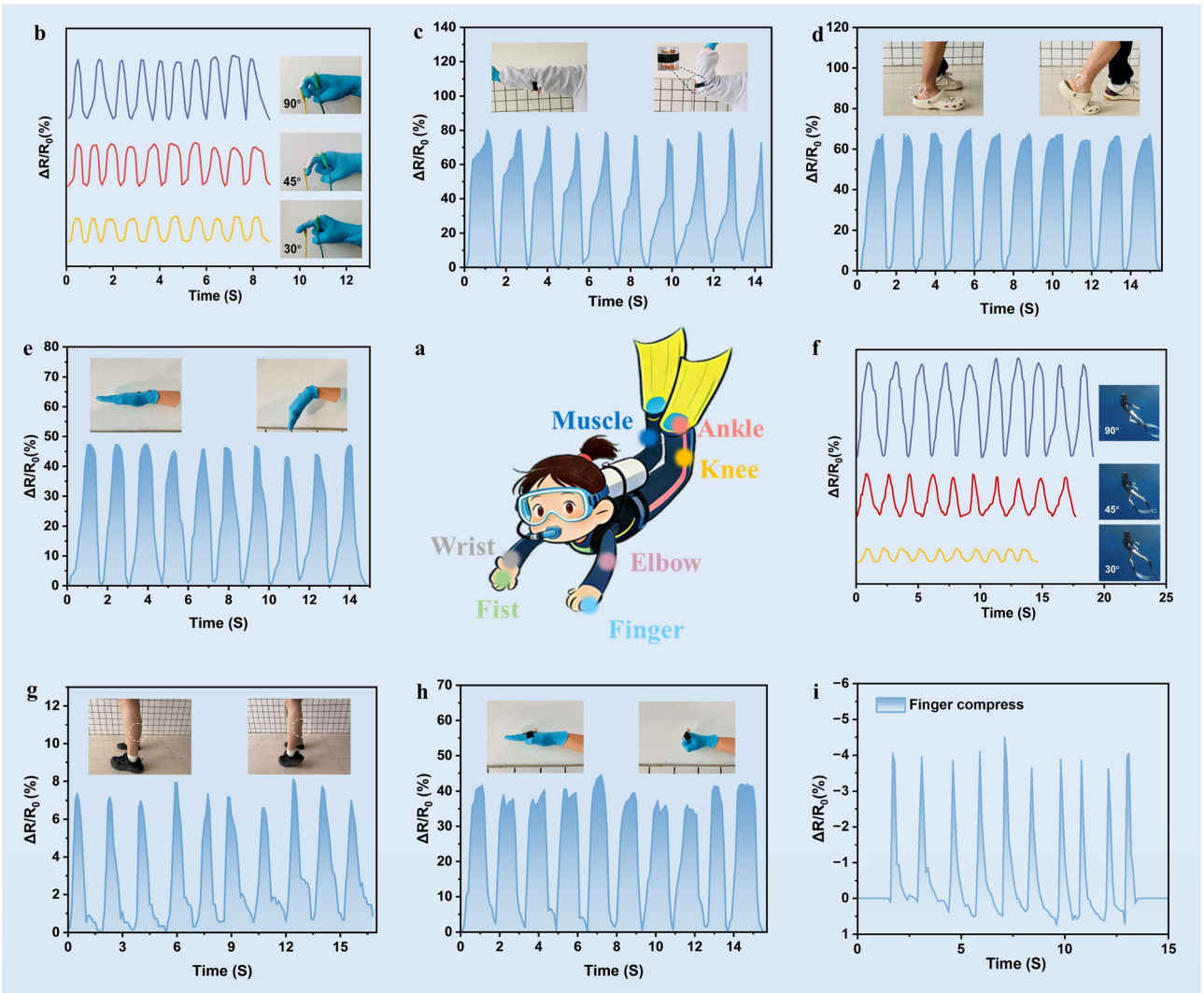


Fig. 8 (a) Schematic diagram showing a diver and the sensor attachment positions. Signal changes detected by the $PANO_{2.5}$@C flexible hydrogel sensor attached to the (b) finger, (c) elbow, (d) ankle, (e) wrist, and (f) knee during (g) muscle exertion, (h) fist clenching, and (i) finger pressing.

monitoring system, providing ground personnel with accurate data analysis and status evaluations (Fig. 9c–e and Fig. S10). Furthermore, the system supports efficient underwater communication based on Morse code principles. Through precise recognition of the pulse duration generated by finger bending movements, "dot" and "dash" signals are distinctly identified. Experimental results demonstrate that individuals equipped with the hydrogel sensor on their fingers can successfully transmit emergency messages, such as "SOS", "HELP", "YES", and "NO", by performing specific bending combinations (Fig. 9f), proving the significant application potential of this multifunctional hydrogel system in smart diving technologies and underwater safety communication.

4. Conclusion

In conclusion, this study successfully developed a composite conductive and thermosensitive hydrogel that combines exceptional mechanical toughness with multimodal sensing capabilities through the copolymerization of AMPS, OMA, and NIPAM, embedded with color-changing microcapsules in a CTAB micelle-confined framework. The hydrogel exhibits not only outstanding mechanical properties (tensile strength 690 kPa; elongation at break 1370%) but also versatile, highly sensitive detection capabilities for temperature (with a sensitivity of −15.344%/°C), human joint movements, and deep-water high-pressure environments (capable of detecting up to 5 MPa, equivalent to a depth of approximately 500 meters). In addition, the material demonstrates remarkable broad-spectrum antibacterial activity, attaining inhibition rates of 99.23% against *E. coli* and 98.7% against *S. aureus*. Inspired by the jellyfish's dynamic environmental adaptability and driven by the synergistic coupling between the internal thermosensitive phase-transition network and the color-changing microcapsules, the hydrogel undergoes distinct color transitions under specific environmental stimuli, thereby providing

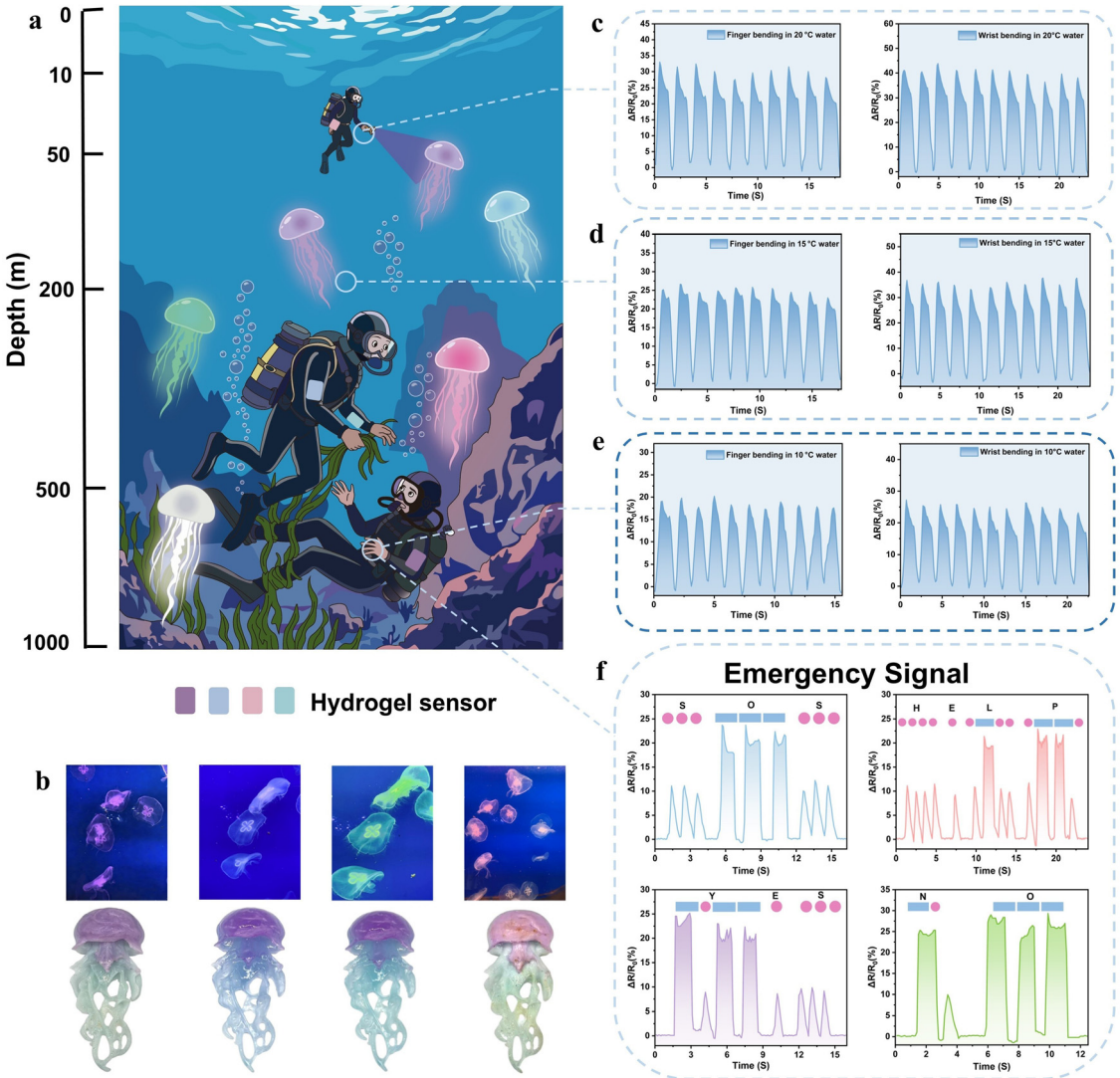


Fig. 9 (a) Schematic illustration of the hydrogel sensor for temperature monitoring and transmission of distress signals from divers in complex underwater environments; (b) physical photos of the color-changing behavior of jellyfish-inspired hydrogels; (c)–(e) resistance change curves of hydrogel for monitoring human motion at different water temperatures; (f) underwater emergency distress signal test based on Morse code for hydrogel, including SOS, HELP, YES, and NO.

intuitive underwater visual warnings. Owing to its structural stability and reliable multimodal sensing performance in complex underwater environments, the hydrogel holds tremendous practical application potential in fields such as underwater communication, emergency rescue scenarios, and smart wearable devices.

Author contributions

Fangrong Sun: writing – original draft, investigation, formal analysis, data curation. Tao Zhang: investigation, data curation, writing – review & editing. Jinmo Yu: investigation, data curation. Aijia Deng: investigation, data curation. Xuan Zhao: investigation, data curation. Bin Fei: investigation, data curation. Katja Loos: investigation, data curation, writing – review & editing. Kunlin Chen: writing – review & editing, supervision, methodology, funding acquisition, conceptualization.

Conflicts of interest

There are no conflicts to declare.

Data availability

Data will be made available on request. The data supporting this article has been included as part of the supplementary information (SI).

Supplementary information (SI) is available. See DOI: https://doi.org/10.1039/d6mh01907b.

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